

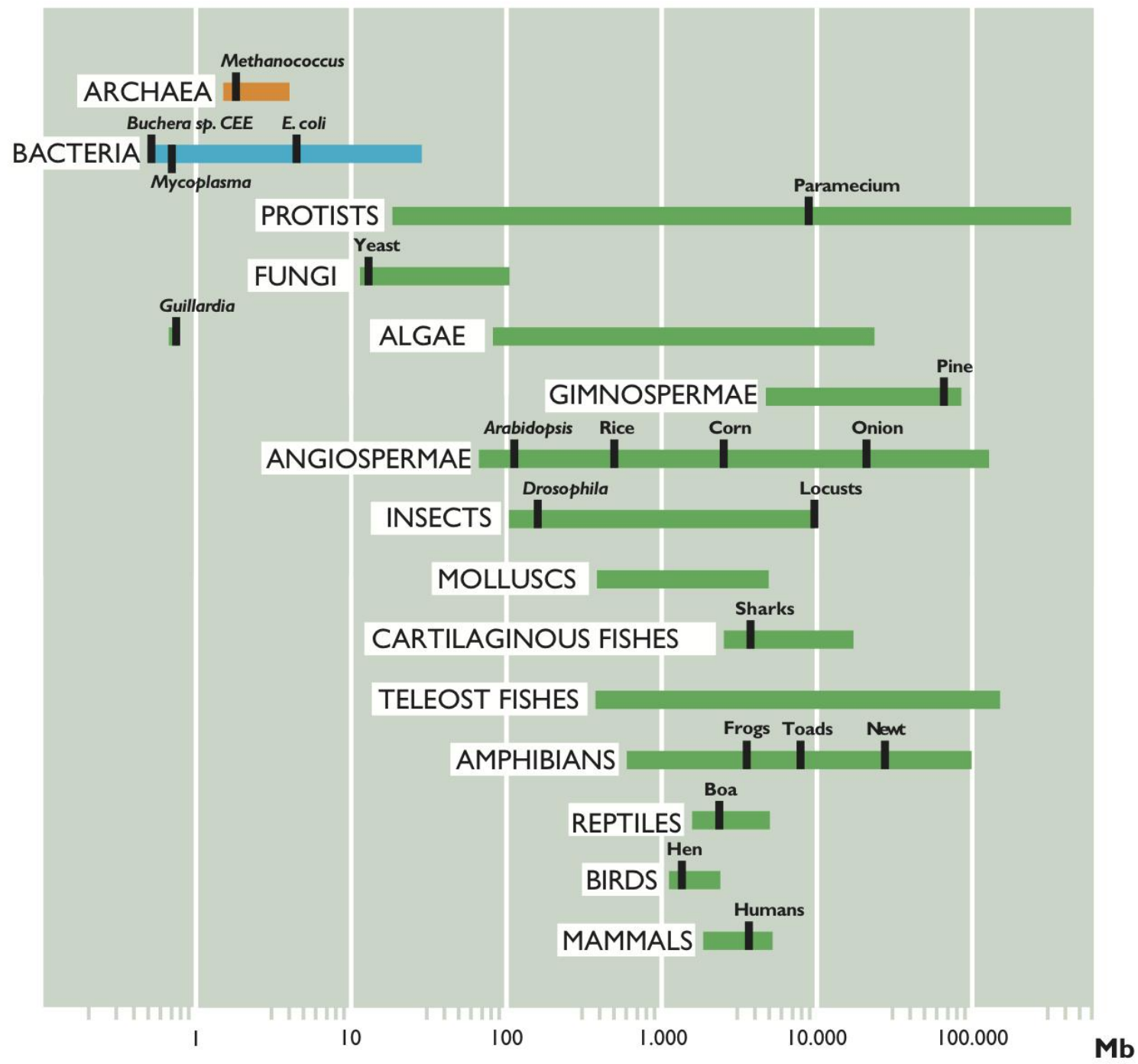


Genome organization

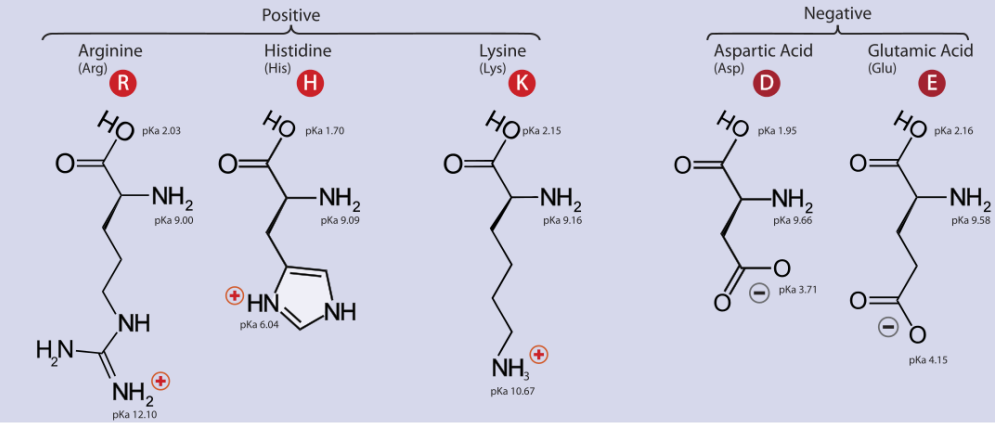
Subha Narayan Rath

Genome, transcriptome, proteomes

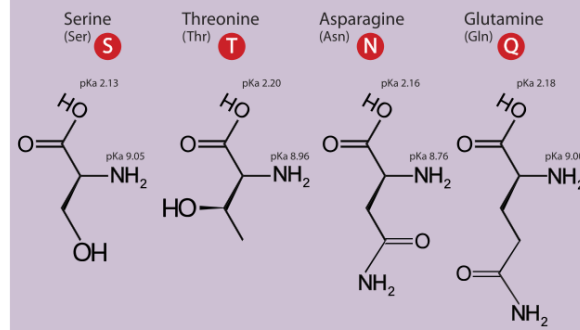
- Genome of a typical bacterium like E. coli: 4.6×10^6 bp with a single DNA molecule
- If extended it would be 2 mm long and it fits into 0.001 mm of diameter of a cell
- Human cells contain 23 pairs of chromosomes and size is 3223×10^6 bp
- Transcriptome: for all the RNA content of the cells
- Proteome: for all the protein content of the cells....in humans only 20000 protein coding genes, but number of proteins are very huge



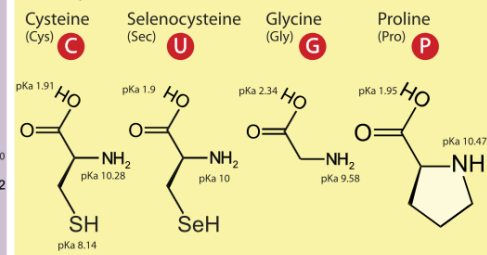
A. Amino Acids with Electrically Charged Side Chains



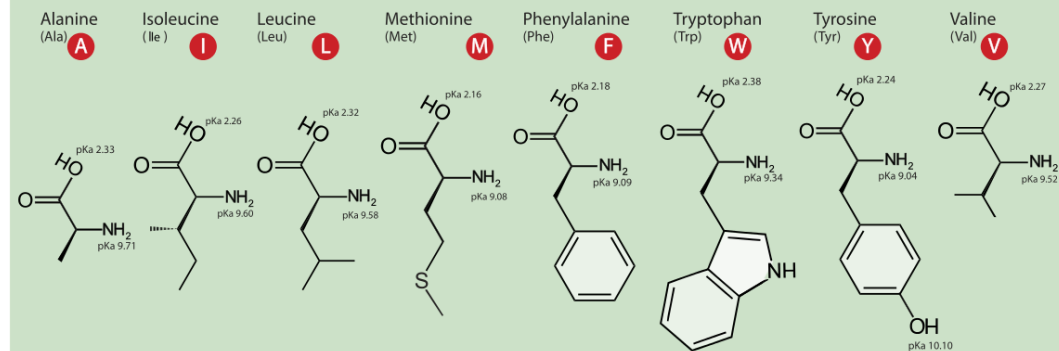
B. Amino Acids with Polar Uncharged Side Chains



C. Special Cases



D. Amino Acids with Hydrophobic Side Chain



Genes

- They may appear in either strand of DNA
- In bacteria: functional unit of genetic sequence are
 - 3N nucleotides presenting
 - N amino acids of a protein
- In eukaryotes: one gene is split into separate segments in genetic DNA
 - EXON: Expressed region
 - INTRON: Intervening region
 - Cellular machinery splices together the initial mRNA to make the product
- Control mechanism may turn genes ON or OFF
 - Or regulate gene expression more finely
 - Cascade of controls respond to conc. of nutrients, to stress, or to control cell cycle

Control regions of DNA lie near the segments coding for proteins

- They contain signal sequences that serve as binding sites for molecules that causes transcription like TFs
- Or, bind regulatory molecules that block transcription
- Bacterial genes: there are OPERONS in line with the genes
- In Eukaryotes: epigenetic signals like
 - DNA methylation
 - Histone modification
- ...they direct tissue specific expression of developmentally regulated genes
- DNA methylation is stable during tissue differentiation surviving cell division
- Reversible chemical modification of histones render the transcription sites more or less accessible
-so it is like 3.2 Gb of data in a mass storage device..

Transcriptomics

- Many RNA transcripts are not protein coding
- Transient: mRNA
- Stable: rRNA
- 1. RNA seq methods by RNA to cDNA and sequencing methods or real time PCR
- 2. RNA can be sequenced directly

BLAST = Basic local alignment search tool

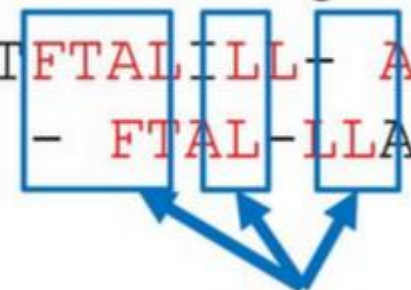
- Rapidly compare a query sequence to a database of subject sequences
- Generate alignments between them= the quality of which is by ALIGNMENT SCORE
- Return alignments that pass user defined score and statistical significance thresholds
- BLAST uses local alignment to find high scoring segment pairs (HSP) between two sequences
- BLAST HIT: A Subject sequence that is aligned to the query

Global Alignment

FTFTALILLAVAV
F - -TAL-LLA-AV

Local Alignment

FTFTALILL- AVAV
- - FTAL-LLAAV--



HSPs

Common blast programs

Program Name	Query	Subject Database
BLAST ^N	Nucleotide	Nucleotide
BLAST ^P	Protein	Protein
BLAST ^X	Nt. → Protein	Protein
^T BLAST ^N	Protein	Nt. → Protein
^T BLAST ^X	Nt. → Protein	Nt. → Protein

BLASTN is for:

- Mapping oligonucleotides to genome
- Comparing DNA from closely related species
- Aligning expressed sequence tags to a genome

BLASTP is for:

- Exploring protein function
- Initial discovery for conserved domains

BLASTX

- Nucleotide query is translated into all 6 reading frames
 - 3 reading frames in + strand
 - 3 reading frames in - strand
- Each reading frame is compared to a protein database
- It is used for:
 - Gene finding in genomic DNA (Annotations)
 - Annotating ESTs

TBLASTN

- Query is a protein sequence
- Nucleotide database is translated into 6 RFs
- The query is then compared to each RF
- It is used for
 - Mapping a protein to genome database
 - Finding ESTs that map to a protein sequence
 - Finding RNA Seq reads that map to a protein sequence

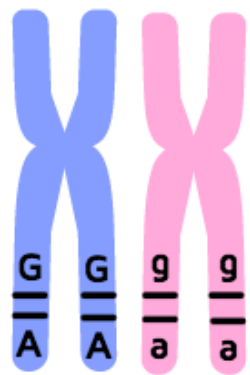
TBLASTX

- BOTH query and subject database (both are nucleotides) are converted to 6 RFs and compared
- It is best used for:
 - Comparing the nucleotide sequence from distantly related species
 - Identify coding regions in ESTs
 - Sensitive but expensive

Transmission of genetic information

- Three types of maps are used:
 - 1. linkage maps of genes
 - 2. banding patterns of chromosomes
 - 3. DNA sequences
- Restriction endonuclease is an enzyme that cuts the DNA at specific regions of around 6bp long...and multiple RE with sequencing gives us overlapping regions for DNA sequencing
- Restriction map: is a map of sizes of fragments stating the order and distance between cleavage sites
- A mutation in one of the sites will change the sizes of fragments allowing the mutation to be located in the map
- Cutting the DNA into smaller pieces, which are cloned and ordered by sequence overlaps produces: contig map (contiguous clone map)

Linked genes
(loci close together)

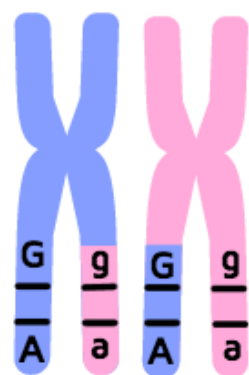


Possible gametes:
GA, ga

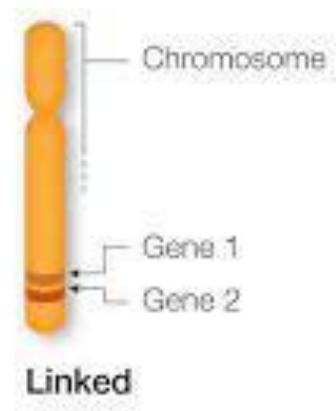
Crossing over during
prophase I of meiosis



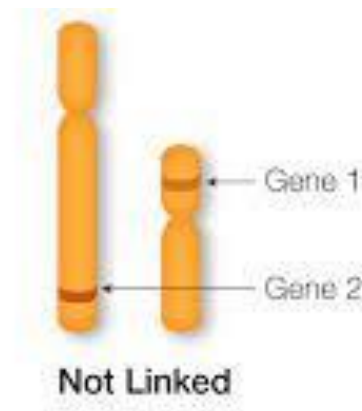
Combination of gametes
remains same as
before cross-over



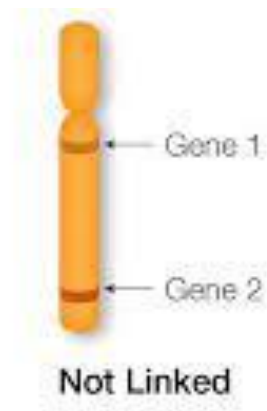
Possible gametes:
GA, ga



Linked



Not Linked



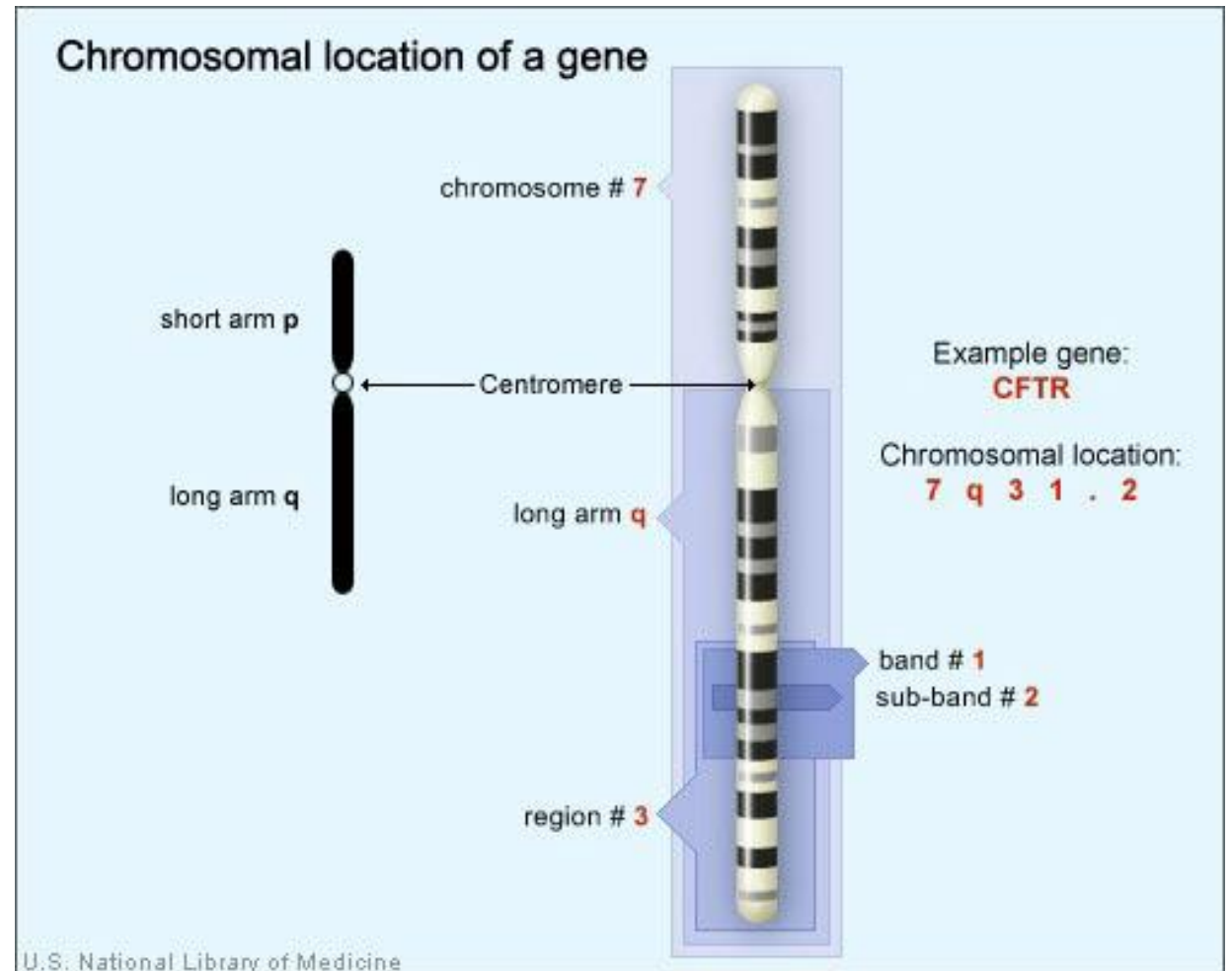
Not Linked

Gene maps, chromosome maps, and sequence maps

- 1. Gene map: observed patterns of heredity
- The farthest apart two linked genes are, the more likely they are to recombine by crossing over during meiosis and considered unlinked
- The unit of length of gene map is Morgan....1cM = 1% recombination frequency = 1×10^6 bp in humans
- Length of gene map = $\frac{\text{new recombinants}}{\text{total offsprings}} \times 100$
- 2. Chromosome banding pattern maps: as chromosomes are physical objects, banding patterns are visible. How nomenclature?
- Numbered in order of size, p for petite arm, p1, p2 etc outward from the centromere, subdivision of bands,
- Deletion of substantial amount of DNA: BY Banding patterns
- Small amount of DNA deletion by: FISH
- 3. DNA SEQUENCE map

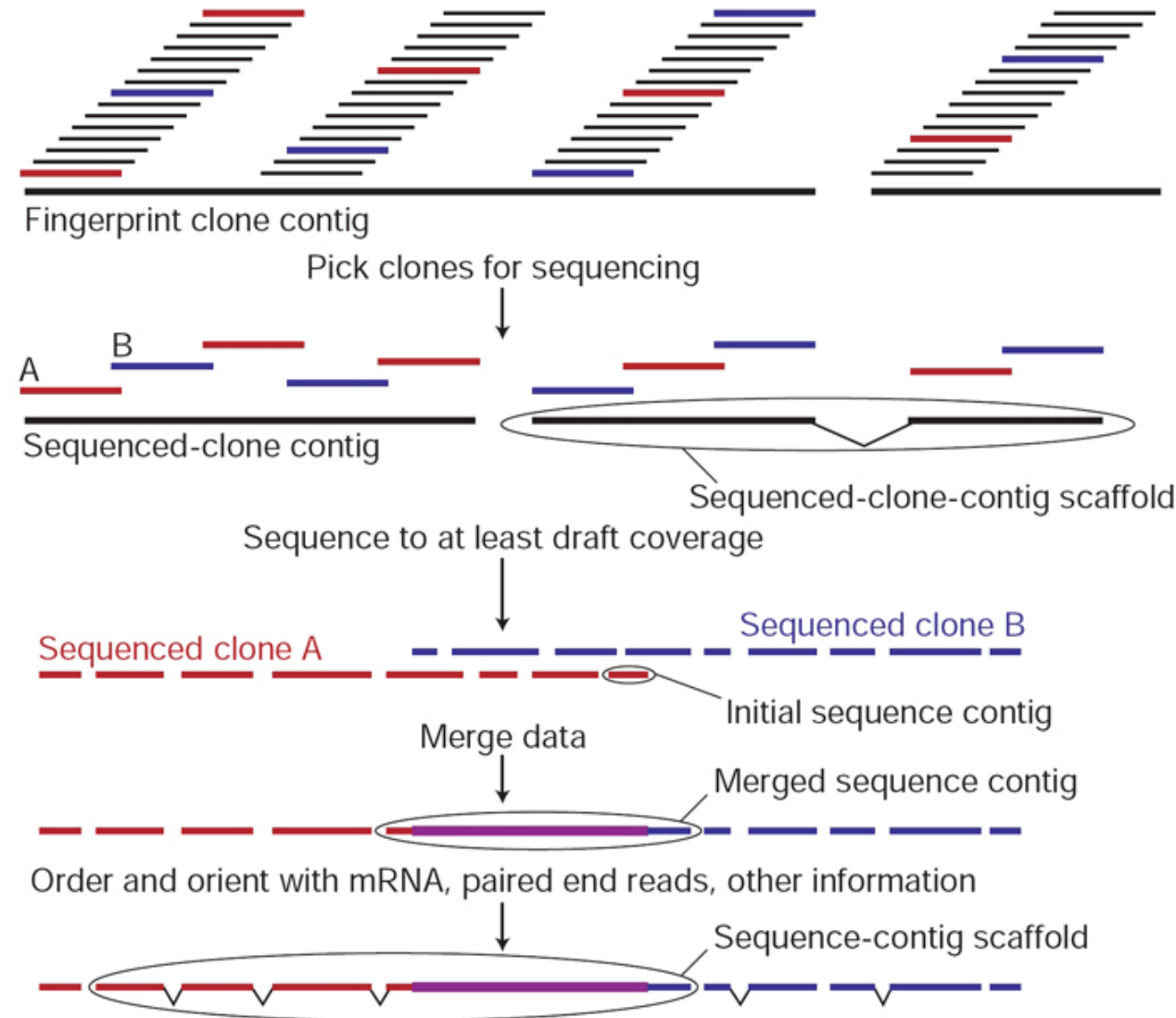
Problem 3.2

- In an insect species, color of the wings and shape of them are linked. On test crossing of the insect at F2, ratio of parental and recombinants are 18:2. what is the distance between the genes in cM?



Picking out genes in genomes

- Computer programs for genome analysis identify open reading frames or ORFs
- ORF is a DNA sequence that starts with an initiator codon, ATG and ends with a stop codon
- An ORF is a potential protein-coding region
- Further approach has 2 ways:
 1. detection of regions similar to known coding regions from other organisms
 2. Ab initio methods to find the genes by the properties of DNA sequence itself



Framework for ab initio gene identification

- Alternate splicing poses a difficulty to find exons and introns
- Initial 5' exon identification
- TATA box ~30bp upstream as core promoter
- Ends before a dinucleotide GT splice signal
- Internal exons like initial exon are free of inframe stop codons
- They begin after AG splice signal and end immediately before a GT splice signal
- Final 3' exon after AG splice signal and ends with a stop codon followed by polyadenylation signal sequence

